

EMET1001 Lecture Week 7

Concave and Convex Functions, Extreme points

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Reading: SHSC chapter 8, 9.1, 9.2

Concave and convex functions

Definition (Concave function)

A function f defined on interval I is said to be concave over I if

$$f(\lambda a + (1 - \lambda)b) \geq \lambda f(a) + (1 - \lambda)f(b)$$

for all a and b in I and all numbers λ in $[0, 1]$.

If the inequality is strict whenever $a \neq b$ and $0 < \lambda < 1$, then f is strictly concave.

Concave and convex functions

Definition (Convex function)

A function f defined on interval I is said to be convex over I if

$$f(\lambda a + (1 - \lambda)b) \leq \lambda f(a) + (1 - \lambda)f(b)$$

for all a and b in I and all numbers λ in $[0, 1]$.

If the inequality is strict whenever $a \neq b$ and $0 < \lambda < 1$, then f is strictly convex.

Concave and convex functions

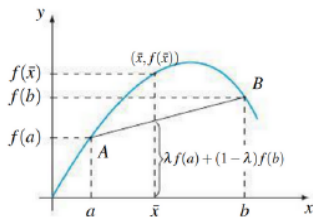


Figure 8.2.1 A concave function

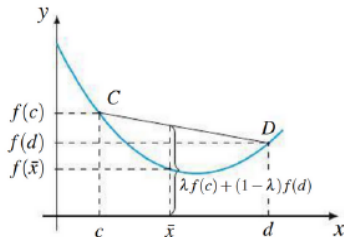


Figure 8.2.2 A convex function

Concave and convex functions slopes

- For twice differentiable functions, we can define concave functions and convex functions by their second order derivatives.
- A positive (negative) second derivative means the first derivative is an increasing (decreasing) function:

$$f''(x) \geq 0 \text{ on } I \quad \Longleftrightarrow \quad f' \text{ is increasing on } I$$

$$f''(x) \leq 0 \text{ on } I \quad \Longleftrightarrow \quad f' \text{ is decreasing on } I$$

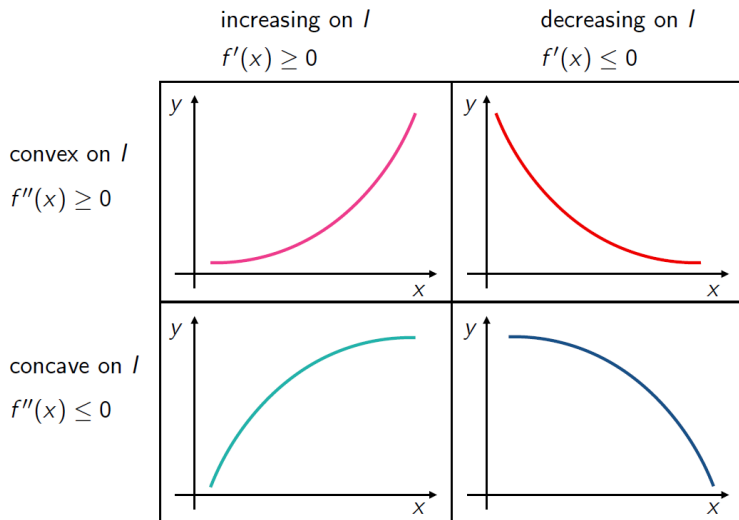
Definition

If $f''(x) \leq 0 \forall x \in I$, then $f(x)$ is concave on I . If $f''(x) \geq 0 \forall x \in I$, $f(x)$ is convex on I .

Concave, convex, and monotonicity

- For functions that are twice differentiable, convexity and monotonicity can be fully characterized in terms of the first and second derivatives:
- A function f defined on interval I :

	$f'(x) \geq 0$	$f'(x) \leq 0$
$f''(x) \geq 0$	$\uparrow$ and convex on I	$\downarrow$ and convex on I
$f''(x) \leq 0$	$\uparrow$ and concave on I	$\downarrow$ and concave on I



Concave, convex, and monotonicity example

$$f(x) = x^2 - 2x + 2$$

First derivative:

Second derivative:

Concave, convex, and monotonicity example

$$f(x) = ax^2 + bx + c$$

First derivative:

Concave, convex, and monotonicity example

$$f(x) = ax^2 + bx + c$$

Second derivative:

Concave, convex, and monotonicity example

$$f(x) = \ln(1 - x^2)$$

Domain

Concave, convex, and monotonicity example

$$f(x) = \ln(1 - x^2)$$

Intersection with x - and y -axis

Concave, convex, and monotonicity example

$$f(x) = \ln(1 - x^2)$$

Increasing/decreasing

Concave, convex, and monotonicity example

$$f(x) = \ln(1 - x^2)$$

Convex/concave

Concave, convex, and monotonicity example

$$f(x) = \ln(1 - x^2)$$

Graph

Sum of concave functions

If $f(x)$ and $g(x)$ are concave functions on the interval I and $h(x) = f(x) + g(x)$, then h is concave on I .

- For any pair of points a and b in I , and any number $\lambda \in [0, 1]$,

$$\begin{aligned} & h(\lambda a + (1 - \lambda)b) \\ &= f(\lambda a + (1 - \lambda)b) + g(\lambda a + (1 - \lambda)b) \end{aligned}$$

- Since both f and g are concave

$$f(\lambda a + (1 - \lambda)b) \geq \lambda f(a) + (1 - \lambda)f(b)$$

and

$$g(\lambda a + (1 - \lambda)b) \geq \lambda g(a) + (1 - \lambda)g(b)$$

Sum of concave functions

This implies

$$\begin{aligned} & h(\lambda a + (1 - \lambda) b) \\ \geq & \lambda f(a) + (1 - \lambda) f(b) + \lambda g(a) + (1 - \lambda) g(b) \\ = & \lambda [f(a) + g(a)] + (1 - \lambda) [f(b) + g(b)] \\ = & \lambda h(a) + (1 - \lambda) h(b) \end{aligned}$$

- If either $f(x)$ or $g(x)$ is strictly concave, then $h(x)$ is strictly concave.
- Sum of convex functions are convex.

Minima of concave functions

If $f(x)$ and $g(x)$ are concave functions on the interval I and $h(x) = \min \{f(x), g(x)\}$, then h is concave on I .

Since both f and g are concave

$$\begin{aligned} f(\lambda a + (1 - \lambda)b) &\geq \lambda f(a) + (1 - \lambda)f(b) \\ &\geq \lambda h(a) + (1 - \lambda)h(b) \end{aligned}$$

and

$$\begin{aligned} g(\lambda a + (1 - \lambda)b) &\geq \lambda g(a) + (1 - \lambda)g(b) \\ &\geq \lambda h(a) + (1 - \lambda)h(b) \end{aligned}$$

This implies

$$\begin{aligned} & h(\lambda a + (1 - \lambda) b) \\ &= \min \{f(\lambda a + (1 - \lambda) b), g(\lambda a + (1 - \lambda) b)\} \\ &\geq \lambda h(a) + (1 - \lambda) h(b) \end{aligned}$$

- Similar argument can be used to prove that if $f(x)$ and $g(x)$ are convex functions on the interval I and $h(x) = \max \{f(x), g(x)\}$, then h is convex on I .

Compositions of concave functions

Let $f(x)$ is a concave function on the interval I . Let $g(x)$ be an increasing and concave function defined over the range of f . Let $h(x) = g(f(x))$, then h is concave.

- Since f is concave

$$f(\lambda a + (1 - \lambda)b) \geq \lambda f(a) + (1 - \lambda)f(b)$$

- Since g is increasing,

$$g(f(\lambda a + (1 - \lambda)b)) \geq g(\lambda f(a) + (1 - \lambda)f(b))$$

- Since g is concave

$$g(\lambda f(a) + (1 - \lambda)f(b)) \geq \lambda g(f(a)) + (1 - \lambda)g(f(b))$$

By definition of h ,

$$h(\lambda a + (1 - \lambda)b) = g(f(\lambda a + (1 - \lambda)b))$$

$$h(a) = g(f(a)) \quad \text{and} \quad h(b) = g(f(b))$$

Compositions of concave functions

We have

$$h(\lambda a + (1 - \lambda) b) \geq \lambda h(a) + (1 - \lambda) h(b)$$

- Similarly, if $f(x)$ is a convex function on the interval I . And $g(x)$ is an increasing and convex function defined over the range of f . Let $h(x) = g(f(x))$, then h is convex.

Inverse of concave functions

If f is increasing and concave on the interval I and $g = f^{-1}$, then g is convex on $f(I)$.

- f concave

$$f(\lambda a + (1 - \lambda)b) \geq \lambda f(a) + (1 - \lambda)f(b)$$

- Let $g(y_a) = a$ and $g(y_b) = b$

$$\begin{aligned} f(\lambda g(y_a) + (1 - \lambda)g(y_b)) &\geq \lambda f(g(y_a)) + (1 - \lambda)f(g(y_b)) \\ &= \lambda y_a + (1 - \lambda)y_b \end{aligned}$$

- Since f is increasing, so is g

$$g(f(\lambda g(y_a) + (1 - \lambda)g(y_b))) \geq g(\lambda y_a + (1 - \lambda)y_b)$$

- Since $g = f^{-1}$

$$\lambda g(y_a) + (1 - \lambda)g(y_b) \geq g(\lambda y_a + (1 - \lambda)y_b)$$

g is convex.

Inverse of concave functions

Similarly, the inverse of an increasing and convex function is increasing and concave.

Supergradient and subgradient

If f is differentiable at a , then the slope $f'(a)$ is called the "gradient" at a .

- If f is concave, then the tangent at a is everywhere above f . Then $f'(a)$ is called a "supergradient" of f at a .

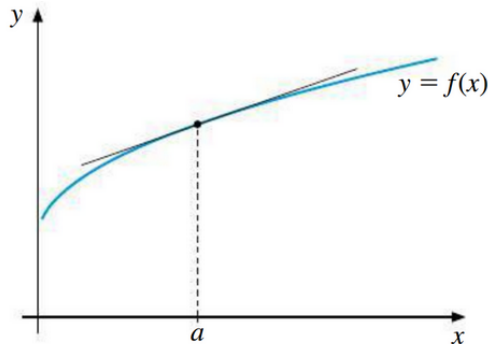


Figure 8.4.1 Supergradient of a concave function

Supergradient and subgradient

Theorem (Sufficient conditions for concavity and convexity)

Suppose that the function f is defined on an interval I . Then f will be:

- ① *concave if it has a supergradient at every point of I ;*
- ② *convex if it has a subgradient at every point of I .*

Inflection points

Points at which a function changes from being convex to being concave, or *vice versa*, are called inflection points.

If f is twice differentiable, a point c is called an inflection point for f if there exists an interval (a, b) about c such that

- ① $f''(x) \geq 0$ in (a, c) and $f''(x) \leq 0$ in (c, b) ; or
- ② $f''(x) \leq 0$ in (a, c) and $f''(x) \geq 0$ in (c, b)

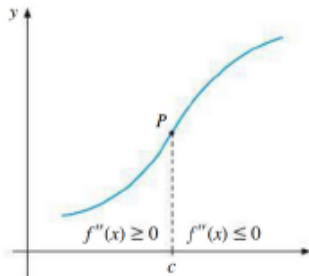


Figure 8.6.1 Point P is an inflection

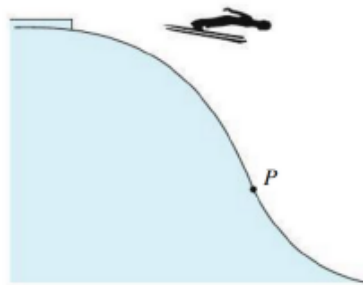


Figure 8.6.2 The point P , where the

Test for inflection points

Let f be a function with a continuous second derivative in an interval I , and let c be an interior point of I .

- 1 If c is an inflection point for f , then $f''(c) = 0$.
- 2 If $f''(c) = 0$ and f'' changes sign at c , then c is an inflection point for f .

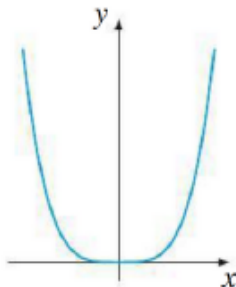


Figure 8.6.3 $y = x^4$

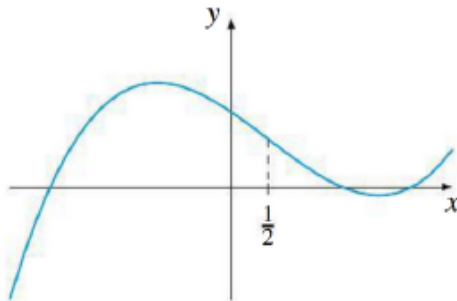


Figure 8.6.4 $y = \frac{1}{9}x^3 - \frac{1}{6}x^2 - \frac{2}{3}x + 1$

Test for inflection points

Test	Inflection point
$f''(x) = 0$ and $f'''(x) < 0$	$\implies$ from convex to concave
$f''(x) = 0$ and $f'''(x) > 0$	$\implies$ from concave to convex

Test for inflection points example

$$f(x) = x^3 - x^2 + 2$$

Test for inflection points example

$$y = x^4$$

Test for inflection points example

$$f(x) = \frac{1}{9}x^3 - \frac{1}{6}x^2 - \frac{2}{3}x + 1$$

Extreme points

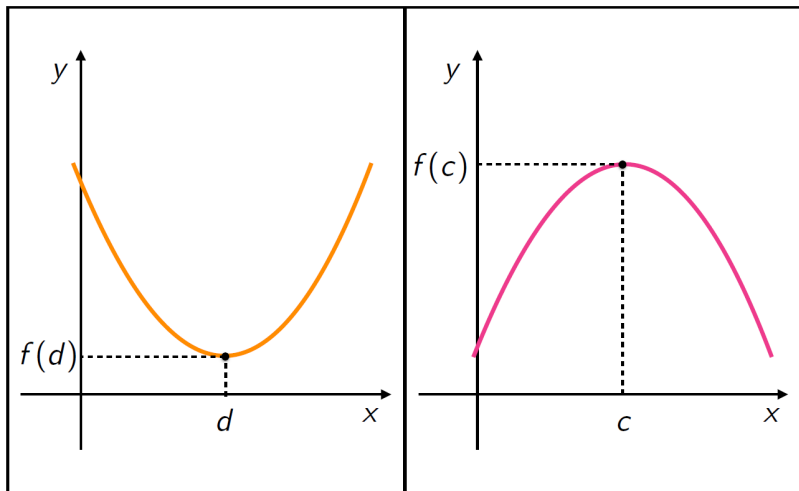
Definition (Global extreme points)

Consider the domain D of the function f , then:

$$\begin{array}{lll} c \in D \text{ is a (global) maximum point} & \iff & f(x) \leq f(c) \quad \forall x \in D \\ d \in D \text{ is a (global) minimum point} & \iff & f(x) \geq f(d) \quad \forall x \in D \end{array}$$

- The maximum value of f is $f(c)$.
- The minimum value of f is $f(d)$.
- An extreme point is either a minimum or a maximum.

Extreme points



Stationary points

Definition (Stationary points)

Suppose that a function f is differentiable in an interval I . A stationary (or critical) point for f is a point c in the interior of I at which

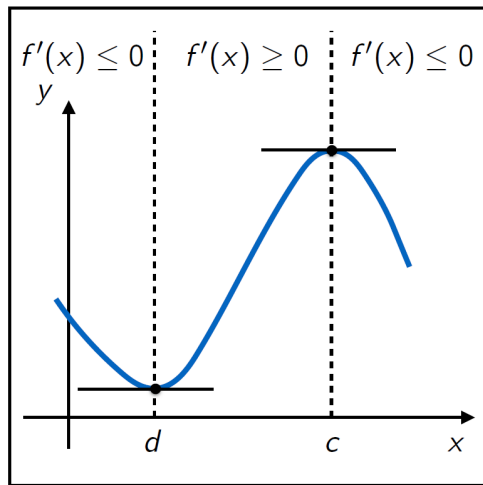
$$f'(c) = 0$$

(Necessary) first-order condition

Suppose that a function f is differentiable in an interval I and that c is an interior point of I . For c to be a maximum or minimum point for f in I , a necessary condition is that it is a stationary point, that is

$$c \text{ is an extreme point for } f \implies f'(c) = 0$$

Tests for extreme points



Extrema of concave and convex functions

Theorem

Suppose that f is a function defined in an interval I and that c is a critical point of f in the interior of I .

- ① *If f is concave, then c is a maximum point for f in I .*
- ② *If f is convex then c is a minimum point for f in I .*

Extrema of strictly concave and strictly convex functions

Theorem

Suppose that f is a function defined in an interval I .

- ① *If f is strictly concave, then any maximum point for f in I is unique.*
- ② *If f is strictly convex then any minimum point for f in I is unique.*

Tests for extreme points example (convex function)

$$f(x) = e^{x-1} - x$$